

Week 6 — Probabilistic Models & Bayesian Inference

Submission Reflection
Statistical Machine Learning · Fusemachines AI Fellowship

Prompt: Give one concrete example where the fully Bayesian answer changed a decision you would have made using only the MLE. Explain the mechanism.

The Decision: Acting on the Small-Segment Churn Rate (Q2 / Q3)

Consider **Group A_small** — a sample of 40 Month-to-month customers drawn at random. The raw MLE gives a churn rate of **37.5%** (15 out of 40 churned). Acting on this point estimate alone, a retention manager might immediately allocate budget for an outreach programme targeting this group, concluding that over a third of the segment is at risk.

The **full Bayesian posterior** — using a Beta(2, 8) prior that encodes domain knowledge that most segments churn below 30% — tells a materially different story. The posterior mode is approximately **33.3%**, and the 94% HDI stretches from roughly **21% to 48%**. The distribution is wide, right-skewed, and its lower bound is barely above the overall dataset churn rate of 26.5%.

The Mechanism

The MLE treats the 40 observations as if they are fully sufficient to characterise the population. It has no memory of all prior customers, no sense that a 15/40 observation is noisy, and it reports a single number with false precision.

The Bayesian posterior does three things the MLE cannot:

- 1. Regularises toward prior knowledge.** With only 40 data points, the likelihood is weak relative to the Beta(2, 8) prior. The prior "pulls" the estimate down by ~4 percentage points — a meaningful correction when the difference between "act now" and "wait for more data" is a few percent.
- 2. Quantifies decision-relevant uncertainty explicitly.** The wide HDI communicates that the true churn rate could plausibly be 21% or 48%. A risk-averse manager reading the posterior distribution would demand a larger sample before committing budget. The MLE gives no such signal.
- 3. Enables direct probability statements.** The Bayesian framework can answer: *"What is the probability that the true churn rate exceeds our intervention threshold of 35%?"* Using the posterior, this is a straightforward integral. The MLE cannot answer this without an asymptotic approximation that is unreliable at $n = 40$.

The Changed Decision

Approach	Estimate	Recommendation
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MLE only	37.5% (point estimate)	Trigger retention campaign immediately — 37.5% is above the 30% company threshold.
Full Bayes	Posterior mean ~33%, mode ~33%, 94% HDI: [21%, 48%]	Collect 200 more observations; HDI width > 10% — decision premature.

Under **Full Bayes**, roughly 25% of the posterior mass sits *below* the 30% threshold. The recommendation becomes: collect 200 more observations, at which point the posterior will be tight enough (HDI width < 10%) to make a confident decision either way. The cost of waiting two weeks for more data is far lower than the cost of a mis-targeted campaign.

Summary

The mechanism is **prior regularisation interacting with small sample size**. When n is small, the MLE variance is high and the single point estimate is a misleading summary of what the data actually say. The Bayesian posterior makes that uncertainty visible and actionable — converting a premature "act now" decision into a calibrated "gather more data" decision.